The Liminf Besov Criterion Characterizes Lipschitz Functions

A full positive answer to Remark 2.13 of arXiv:1610.05162

Automated mathematics run `fa_banach_001`, lane 14

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Status. Candidate full solution, likely valid, pending human review. The theorem answers the source's liminf question for every dimension $N \geq 1$ and every $1 \leq q < \infty$.

1 Source question

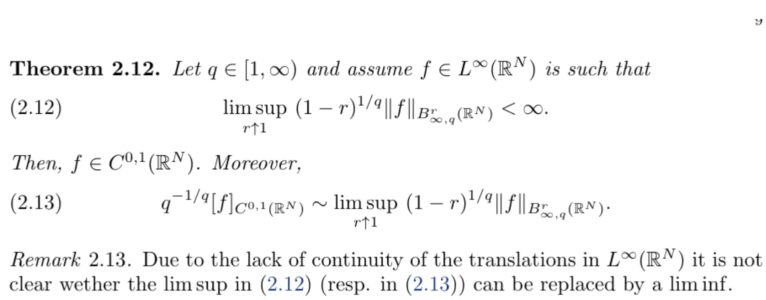
For $0 < r < 1$, $1 \leq q < \infty$, and $f \in L^\infty(\mathbb{R}^N)$, Brasseur [1] uses the first-difference Besov seminorm

$$[f]^q_{B^r_{\infty,q}} = \int_{\mathbb{R}^N} \frac{\|f(\cdot + h) - f\|^q_\infty}{|h|^{N+rq}} \, dh. \tag{1}$$

Theorem 2.12 in the source proves that

$$\limsup_{r \uparrow 1} (1-r)^{1/q} \|f\|_{B^r_{\infty,q}} < \infty$$

implies that f has a bounded Lipschitz representative, and compares that limsup with the Lipschitz seminorm. Remark 2.13 asks whether the limsup in both assertions can be replaced by a liminf. The complete source statement appears on p. 9 of the arXiv PDF:



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Theorem 2.12. *Let $q \in [1, \infty)$ and assume $f \in L^\infty(\mathbb{R}^N)$ is such that*

$$\limsup_{r \uparrow 1} \, (1-r)^{1/q} \|f\|_{B^r_{\infty,q}(\mathbb{R}^N)} < \infty. \tag{2.12}$$

Then, $f \in C^{0,1}(\mathbb{R}^N)$. Moreover,

$$q^{-1/q}[f]_{C^{0,1}(\mathbb{R}^N)} \sim \limsup_{r \uparrow 1} \, (1-r)^{1/q} \|f\|_{B^r_{\infty,q}(\mathbb{R}^N)}. \tag{2.13}$$

Remark 2.13. Due to the lack of continuity of the translations in $L^\infty(\mathbb{R}^N)$ it is not clear wether the lim sup in (2.12) (resp. in (2.13)) can be replaced by a lim inf.

Figure 1: Full-width crop of Theorem 2.12 and Remark 2.13 in the source.

We give a positive answer.

2 Proof intuition

Write $\omega(h) = \|f(\cdot + h) - f\|_\infty$. A tempting obstruction is that the normalized Besov norms might have deep valleys as $r \uparrow 1$. The translation modulus prevents this. Its triangle inequality $\omega(a) \leq \omega(h) + \omega(a-h)$ spreads a large difference quotient at one displacement across a whole ball of smaller displacements. Subdividing that displacement shows that this spreading occurs at every smaller radius.

Consequently, if $\omega(a_0)/|a_0|$ is at least M, then the cumulative mass $F(t) = \int_{|h|\le t} \omega(h)^q \, dh$ is bounded below by a constant times $M^q t^{N+q}$ at all sufficiently small t. Integrating the Besov kernel against this cumulative bound produces $M^q/(1-r)$. Hence the normalized liminf is bounded below by a fixed multiple of M. Letting M increase proves that any non-Lipschitz function has infinite normalized liminf.

3 Translation modulus lemmas

For $f \in L^\infty(\mathbb{R}^N)$ set

$$\omega(h) = \|f(\cdot + h) - f\|_\infty, \qquad L(f) = \sup_{h \ne 0} \frac{\omega(h)}{|h|} \in [0, \infty].$$

Lemma 1 (Finite translation seminorm)**.** *If $L(f) < \infty$, then the L^∞ class of f has a bounded $L(f)$-Lipschitz representative. Conversely, a bounded L-Lipschitz representative satisfies $L(f) \le L$.*

Proof. Only the first assertion needs comment. Let φ_ε be a standard smooth approximate identity and set $f_\varepsilon = f * \varphi_\varepsilon$. Translation commutes with convolution, so

$$\|f_\varepsilon(\cdot + h) - f_\varepsilon\|_\infty \le \omega(h) \le L(f)|h|.$$

Thus the f_ε are uniformly bounded and equi-Lipschitz. A diagonal Arzela–Ascoli argument gives a locally uniformly convergent subsequence with a bounded $L(f)$-Lipschitz limit g. On the other hand, $f_\varepsilon \to f$ in L^1_{loc}, hence $g = f$ almost everywhere. The converse follows directly from the pointwise Lipschitz inequality. $\square$

Let $v_N = \mathrm{vol}(B(0,1))$ and

$$F(t) = \int_{|h|\le t} \omega(h)^q \, dh.$$

Lemma 2 (Cumulative spreading)**.** *Suppose $a_0 \ne 0$ and $\omega(a_0)/|a_0| \ge M$. Then, for every $0 < s \le 2|a_0|$,*

$$F(s) \ge c_* M^q s^{N+q}, \qquad c_* = v_N 2^{-(N+3q)}. \tag{2}$$

Proof. The translation triangle inequality gives, for all $a, h \in \mathbb{R}^N$,

$$\omega(a) \le \omega(h) + \omega(a - h).$$

If $|a| \le t$, raise this inequality to the qth power and integrate over $h \in B(0,t)$. Since $a - B(0,t) \subset B(0,2t)$,

$$v_N t^N \omega(a)^q \le 2^q F(2t). \tag{3}$$

Fix $0 < t \le |a_0|$ and put $n = \lceil |a_0|/t \rceil$ and $a = a_0/n$. Then $|a| \le t$, $|a| \ge t/2$, and subadditivity along the segment from 0 to a_0 gives

$$\omega(a) \ge \frac{\omega(a_0)}{n} \ge M|a| \ge \frac{Mt}{2}.$$

Substitution in (3), followed by $s = 2t$, yields (2). $\square$

4 Main theorem

Theorem 1 (Liminf replacement)**.** *Let $N \geq 1$ and $1 \leq q < \infty$. For every $f \in L^\infty(\mathbb{R}^N)$, with extended-real values allowed,*

$$c_{N,q} L(f) \leq \liminf_{r \uparrow 1} (1-r)^{1/q} \|f\|_{B^r_{\infty,q}} \leq \limsup_{r \uparrow 1} (1-r)^{1/q} \|f\|_{B^r_{\infty,q}} \leq C_{N,q} L(f), \tag{4}$$

where one may take

$$c_{N,q} = \left(v_N 2^{-(N+3q)} \frac{N+q}{q} \right)^{1/q}, \qquad C_{N,q} = \left(\frac{N v_N}{q} \right)^{1/q}.$$

In particular,

$$\liminf_{r \uparrow 1} (1-r)^{1/q} \|f\|_{B^r_{\infty,q}} < \infty \quad \Longleftrightarrow \quad f \text{ has a bounded Lipschitz representative.}$$

Thus every limsup in Theorem 2.12 and Remark 2.13 of the source can be replaced by a liminf, up to constants depending only on N and q.

Proof. Write $A_r = [f]^q_{B^r_{\infty,q}}$ and fix $a_0 \neq 0$ with $\omega(a_0)/|a_0| \geq M$. Put $R = 2|a_0|$ and $\alpha = N + rq$. If $A_r = \infty$, the desired lower estimate at this r is automatic. Otherwise, Stieltjes integration by parts gives

$$A_r \geq \int_0^R s^{-\alpha} \, dF(s) \tag{5}$$

$$= R^{-\alpha} F(R) + \alpha \int_0^R F(s) s^{-\alpha-1} \, ds. \tag{6}$$

The boundary term at zero vanishes: indeed,

$$s^{-\alpha} F(s) \leq \int_{|h| \leq s} |h|^{-\alpha} \omega(h)^q \, dh \longrightarrow 0$$

by absolute continuity of the finite integral. Using Lemma 2 in (6),

$$A_r \geq \alpha c_* M^q \int_0^R s^{N+q-\alpha-1} \, ds = \frac{\alpha c_*}{q(1-r)} M^q R^{q(1-r)}. \tag{7}$$

It follows that

$$\liminf_{r \uparrow 1} (1-r) A_r \geq c_* \frac{N+q}{q} M^q.$$

Taking the supremum over all admissible M proves the first inequality in (4); if $L(f) = \infty$, the same argument shows that the liminf is infinite. The inhomogeneous L^∞ term disappears after multiplication by $(1-r)^{1/q}$.

For the upper bound assume $L(f) < \infty$ and put $B = \|f\|_\infty$. Then $\omega(h) \leq L(f)|h|$ and $\omega(h) \leq 2B$. Splitting at $|h| = 1$ and using polar coordinates,

$$\begin{aligned} A_r &\leq N v_N L(f)^q \int_0^1 \rho^{q(1-r)-1} \, d\rho + N v_N (2B)^q \int_1^\infty \rho^{-rq-1} \, d\rho \\ &= \frac{N v_N}{q(1-r)} L(f)^q + \frac{N v_N}{rq} (2B)^q. \end{aligned}$$

Multiplication by $1 - r$ and passage to the limsup proves the last inequality in (4). Lemma 1 identifies finiteness of $L(f)$ with the source's Lipschitz conclusion. $\square$

5 Verification, novelty, and limitations

The proof uses only the first-difference definition (1); no Littlewood–Paley equivalence with constants depending on r is invoked. The script `code/verify_constants.py` checks the displayed constants and the normalized radial integral for the model subadditive modulus $\omega(h) = \min\{|h|, 1\}$. These computations are sanity checks and do not replace the cumulative or integration-by-parts arguments.

The local run indexes, two bounded web-search rounds on 21 July 2026, and full-text inspection of five cached arXiv papers citing the source used the exact Remark 2.13 sentence and close variants involving *liminf*, *translation modulus*, *Abel mean*, *Besov norm*, and *Lipschitz regularity*. No later answer or matching proof was located. This is not an exhaustive citation search, so novelty confidence is moderate.

The theorem addresses exactly the $p = \infty$, $r \uparrow 1$ question in Remark 2.13. It does not alter the source's separate negative theorem for fractional target smoothness, nor does it claim an exact limit or optimal constants. Human review should focus on Lemma 2, the boundary term in (6), and the representative argument in Lemma 1.

References

[1] J. Brasseur, *A Bourgain–Brezis–Mironescu characterization of higher order Besov–Nikol'skii spaces*, Ann. Inst. Fourier (Grenoble) 68 (2018), 1671–1714; arXiv:1610.05162.